Atoms and Molecules:

- Around 500 B.C, Indian Philosopher named Maharishi Kanad postulated that matter can be divided into extremely small particles, which can't be divided further and are called as Parmanus.

- Greek Philosphers - Democritus and Leucippus gave same idea but called these indivisible particles as "Atoms".

Atom - "That can't be cut".

→ **Laws of Chemical Combination:-**

1) **Law of Conservation of mass:-** (By Antonie Lavoiser)

- During any physical or chemical change the total mass of reactant is equal to total mass of products.

'OR'

"Mass can neither be created nor destroyed".

e.g Mercuric oxide $\xrightarrow{heat}$ mercury + oxygen
100g   92.6g   7.4g

- Total mass of Reactants = Total Mass of Products
100g = 92.6 + 7.4 = 100g

Q) If 10g of Calcium Carbonate ($CaCO_3$) on heating gave 4.4g of carbon dioxide (CO_2) and 5.6g of Calcium Oxide (CaO). Show these observations are in agreement with law of conservation of mass?

Sol: $CaCO_3 \xrightarrow{Heat (\Delta)} CO_2 + CaO$
10g   4.4g + 5.6g

- Acc. to Law of conservation of mass of reactants = Total mass of products.

$$10g = 4.4 + 5.6.$$

- Yes it follows Law of Conservation of mass.

- <u>Law of Constant Composition 'OR' Definite Proportion</u>

☆- Proposed by Joseph Proust.

- A pure chemical compound is always made up of same elements that are combined in same proportion by mass.

e.g Water from any source - River, Lake, Ocean, Rain, etc. is always made up of hydrogen and Oxygen in ratio H_2O

$$2:16 = 1:8.$$

Illy. CO_2 i) $CaCO_3 \xrightarrow{\Delta} CO_2 + CaO$

ii) $C + O_2 \longrightarrow CO_2$

In case (i) and (ii) CO_2 prepared by different method but elements ~~are~~ and ratio is same.

$$CO_2 \rightarrow 12:32$$
$$\Rightarrow 3:8$$